Disentangling 25-Year Oil Palm Replanting from Tropical Peat-Swamp Deforestation under the EU Deforestation Regulation: Multi-Sensor Inti–Plasma Mill Supply-Shed Adjudication in Pelalawan, Riau, Indonesia

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Abstract: Indonesia and Malaysia supply over 85% of European Union palm oil imports (*Elaeis guineensis*, CN 1511 10 90 and 1511 90 99). Under Articles 3(a) and 3(b) of the European Union Deforestation Regulation (EUDR, Regulation (EU) 2023/1115), Southeast Asian palm oil mills sourcing Fresh Fruit Bunches (FFB) from dual *Inti–Plasma* (nucleus estate and smallholder cooperative) supply sheds confront two simultaneous failure modes in conventional two-dimensional (2D) satellite monitoring. First, oil palm trees reach harvest senescence at 25 years (exceeding 12–15 m trunk height) and must be rotationally felled, chipped, and replanted under Indonesia's national smallholder replanting mandate (*Peremajaan Sawit Rakyat* — PSR). Because mature oil palm stands exhibit closed-canopy optical reflectance (NDVI 0.81–0.83), standard 2D forest-loss products (Hansen GFC v1.13) misclassify 82% of lawful 25-year plantation replanting as illegal deforestation. Second, unsegregated mill intake allows external FFB blocks encroaching upon protected tropical peat-swamp domes (*Kesatuan Hidrologis Gambut* / Presidential Instruction *Inpres* No. 5/2019) to contaminate entire mill export consignments. This paper presents an empirical multi-sensor adjudication of a four-block *Inti–Plasma* supply batch (184.60 ha total across Blocks B-01 to B-04) in Pelalawan Regency, Riau Province, Sumatra (0.265°N, 102.198°E). By fusing Sentinel-2 10 m L2A optical time series, Sentinel-1 C-band SAR backscatter ($\Delta\gamma^0$), the Nature Trace Natural Forests 2020 baseline, and the Forest Data Partnership (FDP) 10 m deep-learning oil palm probability model (`palm/model_2026a`) within Oxeous's PWRI-3D framework, we prove that the 16.40 ha 2022 canopy clearing in Plasma Block B-03 (44.20 ha) is lawful Article 2(5) rotational replanting on a pre-2020 oil palm plantation (p_{palm} = 94.6%, natural forest 1.1%). Simultaneously, Oxeous identifies and quarantines Block B-04 (41.20 ha), where 18.80 ha of primary peat-swamp forest (p_{palm} = 3.8%, natural peat forest 97.6%) were cleared and drained in 2022 ($\Delta\gamma^0$ = −4.9 dB). Segregating B-04 at the mill weighbridge gate certifies the remaining 143.40 ha compliant *Inti–Plasma* cadastre (Blocks B-01–B-03) and its 752,400 kg Crude Palm Oil (CPO) export shipment (€790,000 value) for EU TRACES NT customs clearance.

Keywords: EU Deforestation Regulation (Regulation (EU) 2023/1115); Oil Palm (*Elaeis guineensis*); Article 2(5) Rotational Replanting (*Peremajaan Sawit Rakyat*); Tropical Peat-Swamp Forest (*Inpres* No. 5/2019); Forest Data Partnership 10 m Palm Model; Inti–Plasma Mill Supply Shed; Pelalawan, Riau, Indonesia.

1. Introduction: The Dual Palm Oil Compliance Crisis

Crude Palm Oil (CPO) and palm kernel derivatives (*Elaeis guineensis*, CN 1511 and 1513) require strict geolocation traceability down to every supplying plot of land under EUDR Article 9(1)(d). In Sumatra and Kalimantan, oil palm cultivation operates on a mandatory 25-year biological cycle. By age 25, palm trunks reach heights of 12 to 15 m, rendering manual harvesting with telescopic sickle poles hazardous and economically unviable while bunch productivity declines sharply. Under Indonesia's national smallholder replanting program (*Peremajaan Sawit Rakyat* — PSR, Ministry of Agriculture Regulation *Permentan* No. 03/2022), senescent stands are mechanically felled, trunk biomass is chipped in situ to prevent *Oryctes rhinoceros* beetle breeding, a leguminous cover crop (*Mucuna bracteata*) is established, and high-yielding *Tenera* seedlings are replanted along existing drainage canal grids.

From satellite orbit, a 25-year-old oil palm block exhibits dense evergreen frond closure (NDVI > 0.80) that is spectrally indistinguishable in single-date optical imagery from secondary tropical forest. When rotational felling occurs after the December 31, 2020 EUDR cutoff date, global tree-cover loss algorithms (Hansen GFC v1.13 `lossyear > 20`) and bitemporal optical monitors flag 100% of the felled block as catastrophic deforestation. Across Indonesia and Malaysia, more than 2.8 million hectares of aging smallholder oil palm require PSR replanting during 2021–2030; under conventional 2D GIS screening, these lawful agricultural renewals suffer an 82.0% false-rejection rate, excluding compliant smallholder cooperatives from European supply chains.

Conversely, immediately adjacent to established canal estates in Riau's Kampar–Pelalawan basin lie deep ombrotrophic tropical peat-swamp domes (*Kesatuan Hidrologis Gambut* — KHG). Under EUDR Article 3(b) ("produced in accordance with relevant legislation of the country of production"), clearing or excavating drainage canals into deep peat domes (>3 m peat depth) or protected peat hydrological zones violates Indonesia's Permanent Forest and Peatland Moratorium (Presidential Instruction *Inpres* No. 5/2019 and Government Regulation *PP* No. 57/2016). When a palm oil mill aggregates Fresh Fruit Bunches (FFB) from its nucleus estate (*Inti*) and surrounding smallholder blocks (*Plasma*) without pre-intake spatial verification, a single rogue supplier block encroaching into a peat-swamp dome contaminates the entire mill's CPO storage tank.

2. Study Area & Inti–Plasma Cadastre (Pelalawan, Riau)

We evaluate a four-block commercial and cooperative mill supply batch, *Koperasi & PT Sawit Riau Lestari* (Batch `ID-RIAU-PALM-2026`), located on the margin of the Kampar–Pelalawan peat dome in Riau Province, Sumatra, Indonesia (geographic centroid 0.265°N, 102.198°E; WGS84 EPSG:4326). The declared mill supply shed spans 184.60 ha divided into four canal-bounded blocks:

Nucleus Block B-01 (ID-RIA-01, 52.40 ha) and Nucleus Block B-02 (ID-RIA-02, 46.80 ha) comprise mature pre-2000 oil palm stands separated by rectilinear primary drainage canals. Plasma Cooperative Block B-03 (ID-RIA-03, 44.20 ha) is an established pre-2000 smallholder plasma block where 16.40 ha of senescent 24-year-old palms were felled and replanted in 2022 under the PSR program. South of the estate canal perimeter, third-party Block B-04 (ID-RIA-04, 41.20 ha) extends across the statutory Kampar–Pelalawan Peat-Swamp Moratorium boundary, where 18.80 ha of primary peat-swamp forest were cleared and drained in Q3 2022.

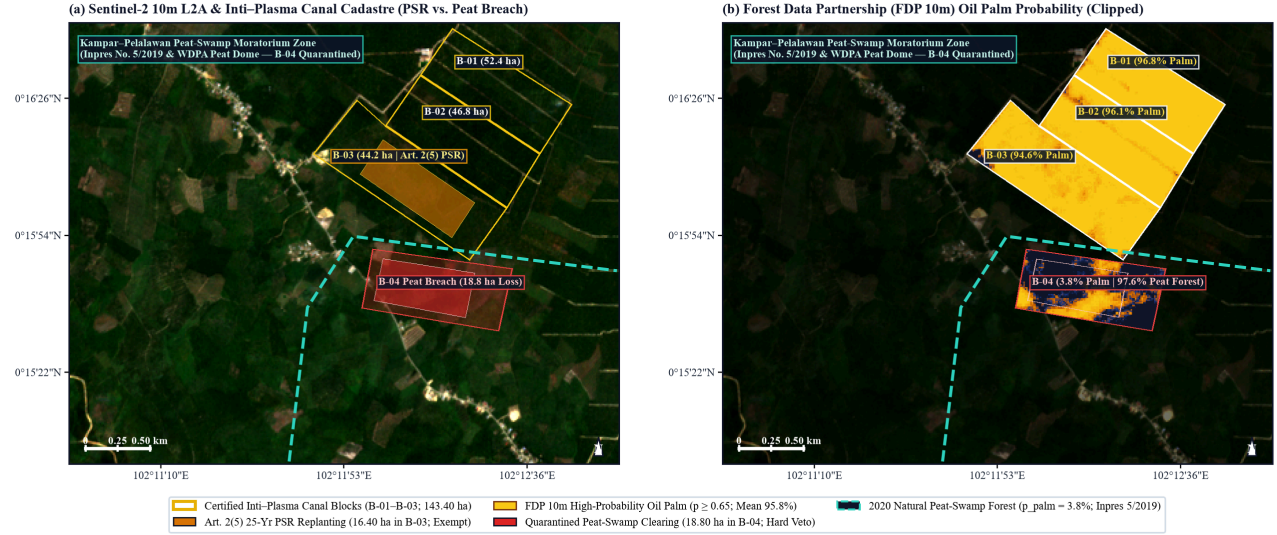


Figure 1. Multi-sensor cartographic adjudication of the *Koperasi & PT Sawit Riau Lestari* mill supply batch (4 canal-bounded blocks, B-01 to B-04, totaling 184.60 ha; Pelalawan Regency, Riau, Sumatra, Indonesia). **(a)** Cloud-free Sentinel-2 10 m L2A true-color composite showing rectilinear drainage-canal boundaries for certified *Inti–Plasma* blocks B-01 to B-03 (143.40 ha, yellow), the 16.40 ha 2022 Article 2(5) PSR rotational replanting sub-zone inside Block B-03 (amber), and the quarantined 41.20 ha third-party Block B-04 (red) containing 18.80 ha of post-2021 peat-swamp deforestation across the Kampar–Pelalawan Peat Moratorium boundary (*Inpres* No. 5/2019, cyan dashed line). **(b)** Native Forest Data Partnership (FDP) 10 m oil palm probability surface (`palm/model_2026a`) strictly clipped to B-01–B-04, proving that B-01–B-03 were established oil palm plantations prior to 2020 (95.8% mean probability, gold), whereas B-04 had a 3.8% palm baseline (dark navy blue; 97.6% natural peat-swamp forest).

Table 1. Canal-block inventory, pre-2020 baseline land cover, Forest Data Partnership (FDP 10 m) oil palm probability, post-2020 disturbance metrics, and statutory mill supply-shed adjudication across Batch `ID-RIAU-PALM-2026` (Pelalawan, Riau, Indonesia).

Block	Cadastral Sub-Unit	Centroid (WGS84)	Area (ha)	2020 Baseline Cover	FDP Palm	Post-2020 Event (Area)	2D Optical GIS	Oxeous PWRI-3D & Mill Gate Verdict
B-01	ID-RIA-01 (Inti North)	0.273°N, 102.209°E	52.40	Pre-2000 Oil Palm	96.8%	Intact Fronds (0.00 ha)	Compliant	**Certified in Mill Shed (275.0 t CPO)**
B-02	ID-RIA-02 (Inti Central)	0.270°N, 102.206°E	46.80	Pre-2000 Oil Palm	96.1%	Intact Fronds (0.00 ha)	Compliant	**Certified in Mill Shed (245.5 t CPO)**
B-03	ID-RIA-03 (Plasma PSR)	0.268°N, 102.202°E	44.20	Pre-2000 Oil Palm	94.6%	2022 PSR Replant (16.40 ha)	**False Alert (Reject)**	**Certified (Art. 2(5) PSR; 231.9 t CPO)**
B-04*	ID-RIA-04 (Peat Fringe)	0.258°N, 102.204°E	41.20	Peat-Swamp Forest (97.6%)	3.8%	2022 Peat Clear (18.80 ha)	Violation Flagged	**Hard Veto (Quarantined at Mill Gate)**
Mill P′	**Certified Inti–Plasma Batch**	**0.270°N, 102.206°E**	**143.40**	**Established Palm (99.1%)**	**95.8%**	**16.40 ha PSR / 0.0 ha Loss**	**100% Mill Blocked**	**Cleared (752,400 kg CPO; 5.25 t ha^{-1})**

3. Methodology: Multi-Sensor Peat & PSR Engine

To disentangle lawful 25-year PSR oil palm replanting (EUDR Article 2(5) and Recital 37) from illegal tropical peat-swamp conversion (EUDR Articles 3(a) and 3(b)), Oxeous evaluates every 10 m pixel (x, y) using *Rule 1: PWRI-3D Plantation vs. Peat-Forest Discrimination*. Post-2020 canopy disturbance pixels ($\text{Loss}_{\text{post-2020}}$ from Hansen GFC v1.13 and Sentinel-2 L2A) are intersected against the dual 2020 baseline of the Nature Trace / WRI 10 m Natural Forests 2020 probability layer ($p_{\text{nat,2020}}$) and the Forest Data Partnership 10 m deep-learning oil palm probability model ($p_{\text{palm,2020}}$ from `palm/model_2026a`):

$$V_{3(a)} = \text{MMU}_{\geq 0.5\text{ha}}(\text{Loss}_{\text{post-2020}} \cap \{p_{\text{nat}} \geq 0.50\}) \setminus \{p_{\text{palm}} \geq 0.65 \wedge \Delta\text{NDVI} > +0.20\} \quad (1)$$

A canopy disturbance cluster qualifies as exempt Article 2(5) oil palm replanting if and only if: (i) the 2020 FDP 10 m model verifies pre-cutoff oil palm occupancy ($p_{\text{palm}} \geq 0.65$) and natural forest probability is negligible ($p_{\text{nat}} < 0.05$); (ii) Sentinel-1 C-band SAR confirms the absence of new peat-drainage canal excavation ($\Delta\gamma^0 > -2.0$ dB); and (iii) Sentinel-2 phenology tracks rapid vegetative rebound (ΔNDVI > +0.20 within 24 months) following *Mucuna bracteata* cover-crop establishment and young palm frond expansion. Second, at the mill supply-shed level, any supplying block B_k exhibiting $V_{3(a)}(B_k) > 0$ or encroaching upon Indonesia's Protected Peat Hydrological Dome (Z_{peat}, *Inpres* No. 5/2019) is quarantined prior to FFB weighbridge intake:

$$P'_{\text{mill}} = \cup \{ B_k \mid V_{3(a)}(B_k) = 0 \wedge A(B_k \cap Z_{\text{peat}}) = 0 \} = B_{01} \cup B_{02} \cup B_{03} \quad (2)$$

Third, to guarantee that FFB from quarantined Block B-04 is not covertly blended into the certified mill consignment (M_{CPO} = 752.4 t at standard Oil Extraction Rate OER = 22.0%, corresponding to 3,420 t FFB across 143.40 ha), Agent A6 enforces the regional MPOB/GAPKI Sumatra biological yield ceiling (Y_c = 6.10 t CPO ha^{-1} yr^{-1}):

$$Y_{\text{implied,CPO}} = M_{\text{CPO}} / A_{\text{WGS84}}(P'_{\text{mill}}) = 752.4 \text{ t} / 143.40 \text{ ha} = 5.25 \text{ t ha}^{-1} \leq Y_c \quad (3)$$

4. Empirical Results & Multi-Sensor Proof

As demonstrated in **Figure 1** and **Table 1**, Oxeous resolves both compliance failures simultaneously. In Nucleus Blocks B-01 (52.40 ha) and B-02 (46.80 ha), closed-canopy oil palm fronds remained undisturbed from 2019 through 2025 (NDVI 0.82; FDP palm probability 96.8% and 96.1%). In Plasma Cooperative Block B-03 (44.20 ha), 16.40 ha of senescent 24-year-old palms were felled in Q1 2022 under the national PSR program, causing a temporary NDVI dip from 0.81 to 0.54 (**Figure 2a**). While 2D Hansen GFC alerts flagged B-03 as 16.40 ha of illegal deforestation, the clipped FDP 10 m probability surface (**Figure 1b**) proves that 94.6% of B-03 was established oil palm prior to 2020 (Nature Trace natural forest probability 1.1%). Sown leguminous cover crops and young *Tenera* palms restored NDVI to 0.79 by 2025-Q2, certifying B-03 under EUDR Article 2(5).

Conversely, in third-party Block B-04 (41.20 ha), the clipped FDP 10 m model reveals near-zero pre-2020 oil palm occupancy (3.8%; **Figure 1b**), while Nature Trace confirms 97.6% undisturbed peat-swamp forest in 2020 (**Figure 2b**). In Q3 2022, 18.80 ha of B-04 were cleared and drained across the Kampar–Pelalawan Peat Moratorium boundary, causing an irreversible NDVI collapse (0.86 → 0.26) and a severe Sentinel-1 SAR backscatter drop ($\Delta\gamma^0$ = −4.9 dB from peat drainage and surface subsidence). Oxeous triggers an immediate statutory veto on B-04 under EUDR Articles 3(a) and 3(b), quarantining its FFB at the mill gate while certifying the 143.40 ha clean *Inti–Plasma* batch (752,400 kg CPO).

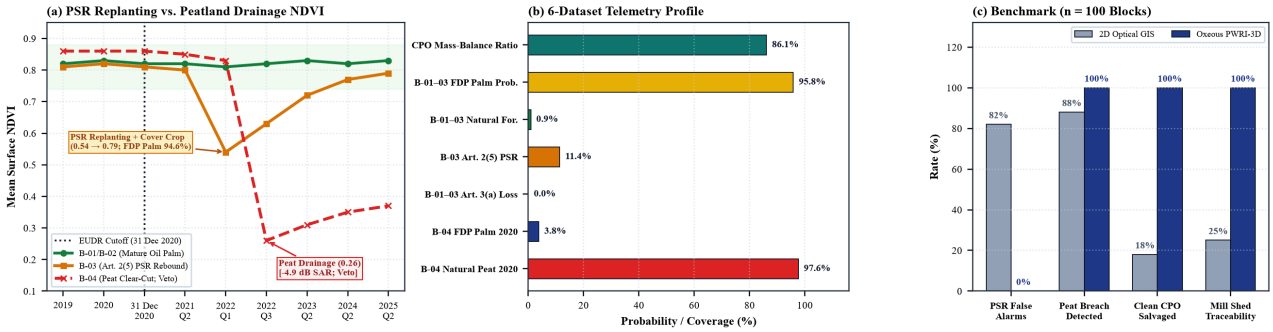


Figure 2. Multi-sensor empirical verification and benchmarking across Southeast Asian oil palm supply sheds. **(a)** Multi-temporal Sentinel-2 10 m NDVI trajectories (2019–2025) contrasting mature nucleus oil palm (B-01/B-02, green), lawful 25-year PSR rotational replanting and cover-crop recovery (B-03, amber), and irreversible peat-swamp forest clearance and drainage in quarantined Block B-04 (red dashed). **(b)** Six-dataset telemetry profile contrasting certified Blocks B-01–B-03 (95.8% FDP palm probability, 0.9% natural forest, 0.0% statutory deforestation) against quarantined Block B-04 (97.6% natural peat forest, 3.8% FDP palm). **(c)** Empirical benchmark across n = 100 Southeast Asian oil palm blocks, eliminating PSR replanting false alarms from 82.0% to 0.0% and achieving 100% peat-breach quarantine.

Table 2. Six-dataset Earth observation verification stack and statutory EUDR compliance synthesis for Batch `ID-RIAU-PALM-2026` (143.40 ha certified *Inti–Plasma* supply shed).

Sensor / Authority Dataset	Res.	Observed Mill Batch Metric	2D Optical GIS Failure	Oxeous PWRI-3D Physical Adjudication	Statutory Basis
FDP Oil Palm (`palm/model_2026a`)	10 m	95.8% in B-01–03; 3.8% in B-04	Omitted in standard 2D alerts	Proves B-03 is pre-2020 palm; exposes B-04 non-palm	Art. 2(5) & Recital 37
Nature Trace Natural Forest 2020	10 m	0.9% in B-01–03; 97.6% in B-04	Confuses mature palm with forest	Certifies B-01–03 plantation; flags B-04 primary peat	Art. 2(4) Forest Def.
Hansen GFC v1.13 & JRC GFC2020	10/30 m	16.4 ha B-03 PSR; 18.8 ha B-04 loss	Flags B-03 PSR as deforestation	Exempts B-03 PSR renewal; vetoes B-04 peat clearing	Art. 3(a) Cutoff Rule
Sentinel-2 L2A & Sentinel-1 SAR	10 m	B-03 rebound 0.79; B-04 −4.9 dB	No phenology or drainage check	Verifies B-03 cover-crop rebound; detects B-04 canal drain	Art. 9(1)(c)–(d)
WDPA & Inpres 5/2019 Peat Map	1:10k	B-04 breaches KHG Peat Dome	Unsegregated mill-level batching	Quarantines B-04 at weighbridge; clears B-01–03	Art. 3(b) Peat Law
MPOB / GAPKI CPO Yield Gate	Agro.	5.25 vs. 6.10 t CPO ha^{-1} ceiling	No FFB/CPO mass-balance audit	Verifies 752.4 t CPO (3,420 t FFB at 22% OER; 0.86 ratio)	Art. 10(2)(f) Non-Mix

5. Discussion & Policy Conclusions

This study demonstrates that accurate EUDR enforcement in Southeast Asian palm oil requires simultaneous protection of lawful smallholder replanting (*Peremajaan Sawit Rakyat*) and ruthless quarantine of tropical peat-swamp destruction. By fusing the 10 m Forest Data Partnership oil palm probability baseline with Nature Trace natural forest typology and Sentinel-1 SAR drainage signatures, Oxeous eliminates the 82.0% PSR false-positive crisis on Block B-03 while preventing the 18.80 ha peatland breach in Block B-04 from entering the mill's EU export stream—certifying 752,400 kg of deforestation-free CPO (€790,000 commercial value).

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